

Auteur: Tala Mohanna
Studierichting: Informatica
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$$P(z) = 3z^3 + z^2 + 12z + 4$$

$$= (3z^3 + z^2) + (12z + 4)$$

$$= z^2(3z + 1) + 4(3z + 1)$$

$$= (3z + 1)(z^2 + 4)$$

$$3z + 1 = 0 \quad \Rightarrow \quad z = -\frac{1}{3}$$

$$z^2 + 4 = 0 \quad \Rightarrow \quad z^2 = -4 \quad \Rightarrow \quad z = \pm 2i$$

$$P(z) = (3z + 1)(z^2 + 4)$$

$$z = -\frac{1}{3}, z = 2i, z = -2i$$

References